Particulars of the Experiments Performed

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FLOW PROCESS CHART

Existing Method / Proposed Method

SUMMARY

ACTIVITY	PRESENT	PROPOSED	SAVINGS
OPERATION	8	5	3
TRANSPORTATION	3	3	0
INSPECTION	2	1	1
DELAY	0	0	0
STORAGE	0	0	0
DISTANCE TRAVELED	13	9	4

JOB Aluminium Foil mlf

☐ MAN OR ☑ MATERIAL

CHART BEGINS Transportation of metal

CHART ENDS Cutting

CHARTED BY Sudhanva S. DATE 19/11/25

	DETAILS OF METHOD	OPERATION ○	TRANSPORT ⇨	DELAY D	INSPECTION □	STORAGE ▽	DISTANCE IN METERS	QUANTITY	TIME	REMARKS
1.	Transportation of Al. block									
2.	Melting of Metal									
3.	Sample inspection									
4.	Transportation of tapping melt									
5.	Impurities filtered									
6.	Cooling of metal									
7.	Transportation of rolling m/c									
8.	Thining of metal									
9.	Inspection of metal width									
10.	Calibrating of thin metal									
11.	Cutting of damaged edges									
12.	Surface finishing									
13.	cutting desired length									
14.										
15.										

Signature of the Staff in-charge

FLOW PROCESS CHART

Existing Method / Proposed Method

SUMMARY

ACTIVITY	PRESENT	PROPOSED	SAVINGS
OPERATION	8	5	3
TRANSPORTATION	3	3	0
INSPECTION	2	1	1
DELAY	0	0	0
STORAGE	0	0	0
DISTANCE TRAVELED	13	9	4

JOB Aluminium foil mlf

☐ MAN OR ☑ MATERIAL

CHART BEGINS Transportation of metal

CHART ENDS Cutting to desired length

CHARTED BY Sudhanva S DATE 19/11/25

	DETAILS OF METHOD	OPERATION ○	TRANSPORT ⇨	DELAY D	INSPECTION □	STORAGE ▽	DISTANCE IN METERS	QUANTITY	TIME	REMARKS
1	Transportation of metal									
2	Melting									
3	Inspection									
4	Transportation									
5	Filtering									
6	Cooling of metal									
7	Transportation to rolling m/c									
8	Thining of metal sheet									
9	Cutting to desired length									
10										

Inference:

> Flow process chart (material type) records how the material is handled by the operator.

> It makes use of 5 symbols accounting for operation, transport, delay, inspection performed by workers.

> Flow process chart records man into man operational process chart it evaluates all the activities performed by the worker.

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MULTIPLE ACTIVITY CHART

Existing Method / Proposed Method

Name: Sudhanva Seetha **Reg. No. :** RVCE25BIM405 **Semester:** 3rd

Expt. No. : M302 **Date:** 29/10/25

Activity	Compaction test on sand	Chart No.		Product No.		Charted By	Sudhanva Seetha
Resources	Operators, weighing m/c, Ramming, UTM	Sheet No.		Drawing No.		Date	29/10/25

SUMMARY

Resources	Resource 1: Workers			Resource 2: Weighing m/c			Resource 3: UTM, Raining m/c		
Method	Present	Proposed	Savings	Present	Proposed	Savings	Present	Proposed	Savings
Working Time	249	146	103	58	47	11	64	49	15
Idle Time	0	0	0	191	99	92	185	97	88
Utilization %	100	100	100	23	33	78.67	25.70	33.56	14.56
Cycle Time	249	146	103	Remarks	146	103	249	146	103

Resource 1:		Resource 2:		Resource 3:	
Activity Present Method	Time in CMS	Activity Proposed Method	Time in CMS	Activity	Time in CMS
Weigh the sand with moisture	58	Weigh the sand with moisture	47		
Mixing the sand	47	Mixing & moving the sand	17		
Move to Ramming m/c	20	Preparing the moulds	15		
Preparing the moulds	35	Ramming	19		
Ramming	34	Removal of the specimen	8		
Removal of the specimen	15	Move to UTM	10		
Move to UTM	10	operation	20		
operation, inspection	20,10	inspection	10		

Signature of the Staff in-charge

Summary of % Utilization

Resources	workers	Welding m/c	VTM & Forming m/c
Existing method	$\frac{249}{249} \times 100 = 100\%$	$\frac{58}{249} \times 100 = 23.29\%$	$\frac{64}{249} \times 100 = 25.70\%$
Proposed method	$\frac{146}{146} \times 100 = 100\%$	$\frac{47}{146} \times 100 = 32.19\%$	$\frac{49}{146} \times 100 = 33.56\%$

Inference:

* Multiple activity chart is used to track several participants working together either sequently or simultaneously to produce a work cycle.
* The utilization of each resource is numerically calculated & analyzed by evaluating the time of each activity in cms.
* The analysis is used to improve the existing method such that cycle is reduced and utilization of each resource is minimized.

Parameters	Max. Marks	Marks Obtained	Signature of Staff in-charge
Conceptual Understanding	05		
Conduction of the experiment	06		
Analysis, Results and Inference	06		
Viva	03		
Total	20		

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TWO HANDED PROCESS CHART

SUMMARY

Present Method ✓ / Proposed Method

NAME : Sudhanva · Srivka
EXPERIMENT: Pin board assembly M303

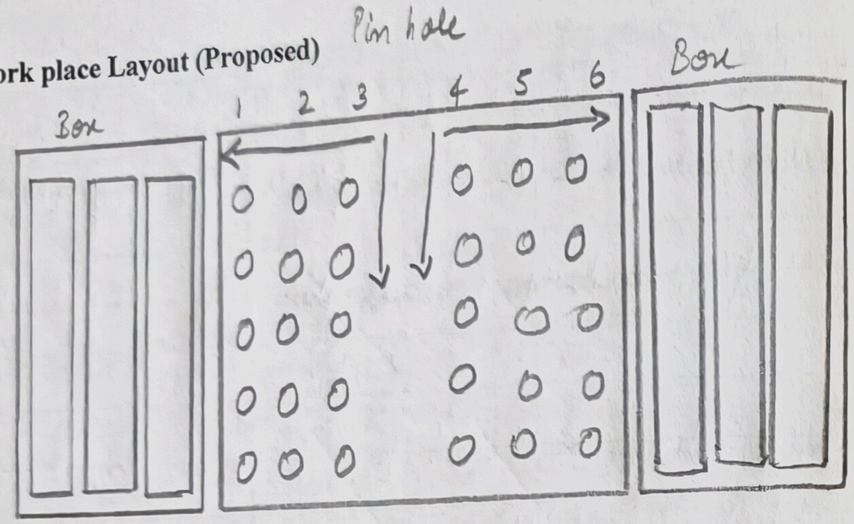
REG. NO: RUCE25BTAP101
DATE : 29/10/25

ACTIVITY	PRESENT		PROPOSED		SAVINGS	
	LH	RH	LH	RH	LH	RH
○ OPERATION	0	7	3	3	0	4
⇨ TRANSPORTATION	2	6	3	3	0	3
D DELAY [Idle]	3	3	4	4	0	0
▽ HOLD	11	0	0	0	11	0

Drawing & Part ..Pin board.....
Operation: assembly of pin board
Location: WSD Lab
Operative: .....
Charted By: Sudhanva S Date: 29/10/25

Left-Hand Description	OPERATION	TRANSPORT	DELAY	HOLD	OPERATION	TRANSPORT	DELAY	HOLD	*Right- Hand description*
Idle									Idle
Picking up the pins									Idle
Hold									Moving the pin
Hold									Placing
Hold									Moving
Hold									Placing
Hold									Moving
Hold									Placing
Idle									Idle
Picking the pins									Placing
Hold									Moving
Hold									Placing
Hold									Moving
Hold									Placing
Hold									Moving
Hold									Placing
Idle									

Work place Layout (Proposed)



Procedure - Proposed Method:

* Place the pinbox on either side of the board to reduce hand movements.
* Pickup 3 pins in each hand
* Move left hand towards 3rd column is 1st row: right hand towards 4th column is 1st row.
* Place the pins with both hands simultenously such that left hand inserts pins in column 1, 2, 3 and right hand in column 4,5,6.
* Repeat the exact same procedure for remaining 4 rows.
* completion of the task with above method reduces the time taken for completion.

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TWO HANDED PROCESS CHART

SUMMARY ✓
Present Method / Proposed Method

NAME : Sudhanva. Sirsa
EXPERIMENT: Pin board assembly

REG. NO: RVCE25BIM405
DATE : 29/10/25

ACTIVITY	PRESENT		PROPOSED		SAVINGS	
	LH	RH	LH	RH	LH	RH
○ OPERATION	0	7	3	3	0	4
⇨ TRANSPORTATION	2	6	3	3	0	3
D DELAY [Idle]	3	3	4	4	0	0
▽ HOLD	11	0	0	0	11	0

Drawing & Part : Pin board
Operation: Assembly of pin board
Location: IISD Lab
Operative:
Charted By: Sudhanva. S Date: 29/10/25

Left-Hand Description	○ OPERATION	⇨ TRANSPORT	D DELAY	▽ HOLD	○ OPERATION	⇨ TRANSPORT	D DELAY	▽ HOLD	*Right- Hand description*
Idle									Idle
Moving									Moving
Placing									Placing
Idle									Idle
Moving									Moving
Placing									Placing
Idle									Idle
Moving									Moving
Placing									Placing

Signature of the Staff in-charge

Inference

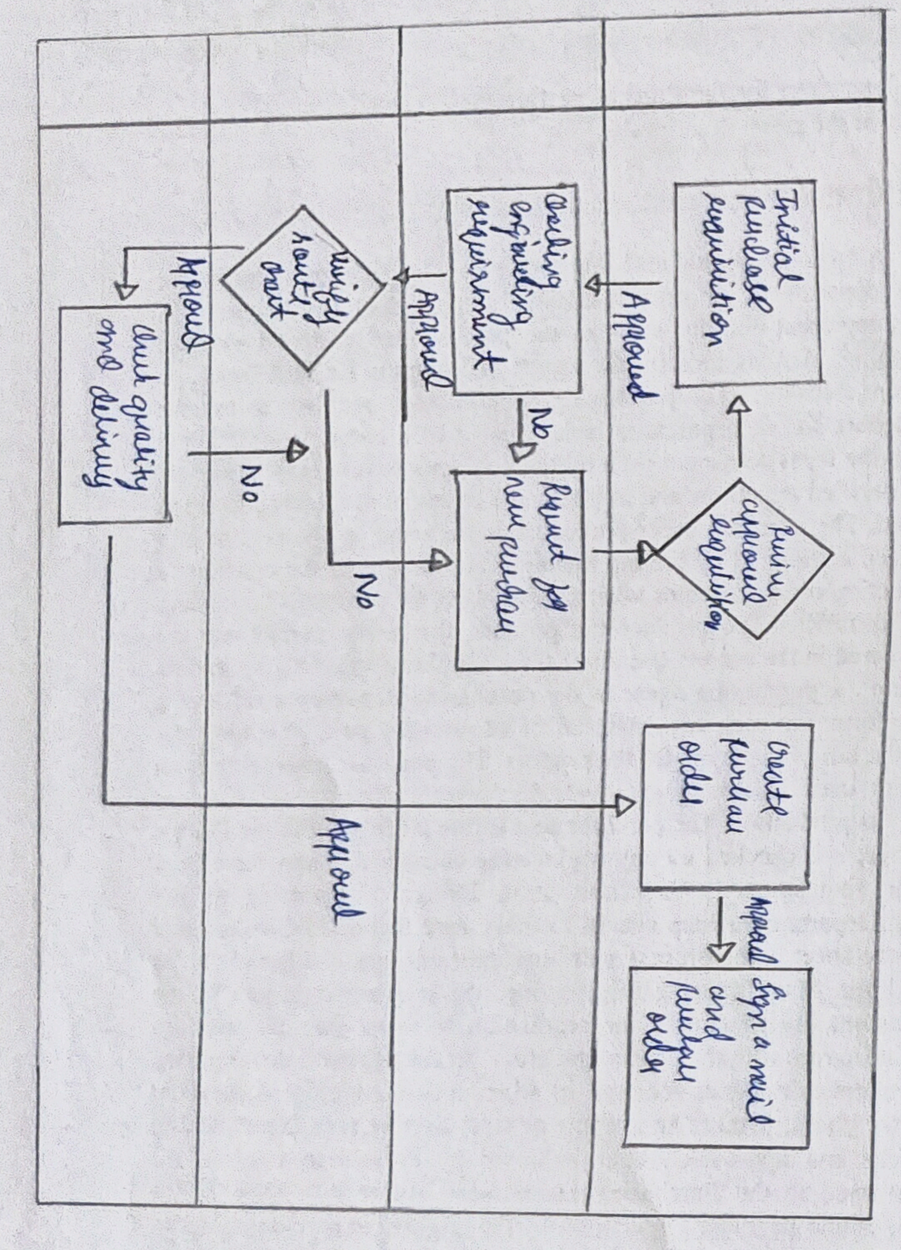
* The two handed process chart is used to track any activities of movements done by the left hand and right hand of operator.

* It records defined categories such as operation, transportation, delay and hold.

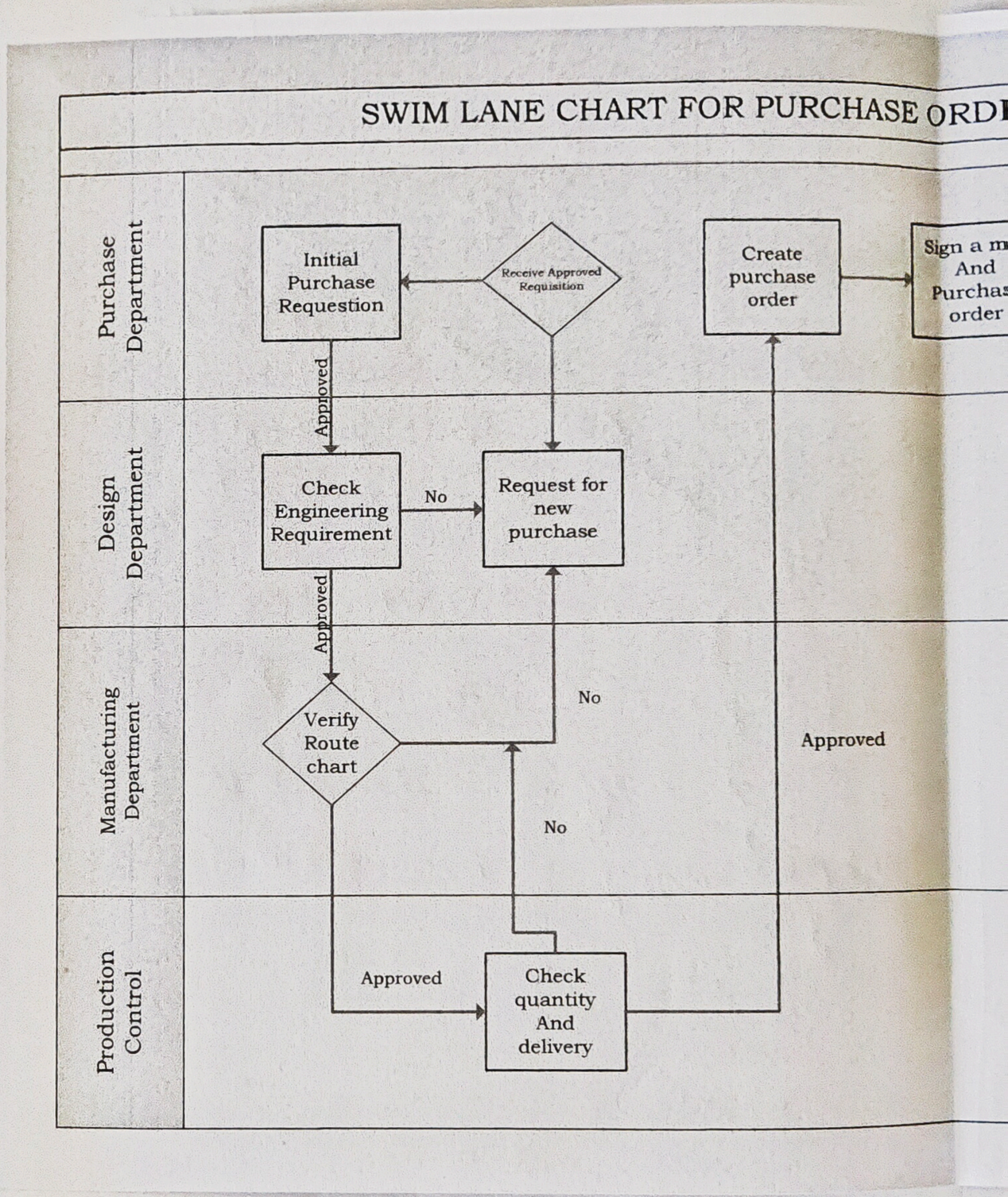
* In the present method, the hands were not operated simultaneously hence leading to a longer time in completing the task.

* In the proposed method, the hands are used simultaneously thereby significantly reducing the time taken to complete the task.

Space for Inserting the Chart (Manual):

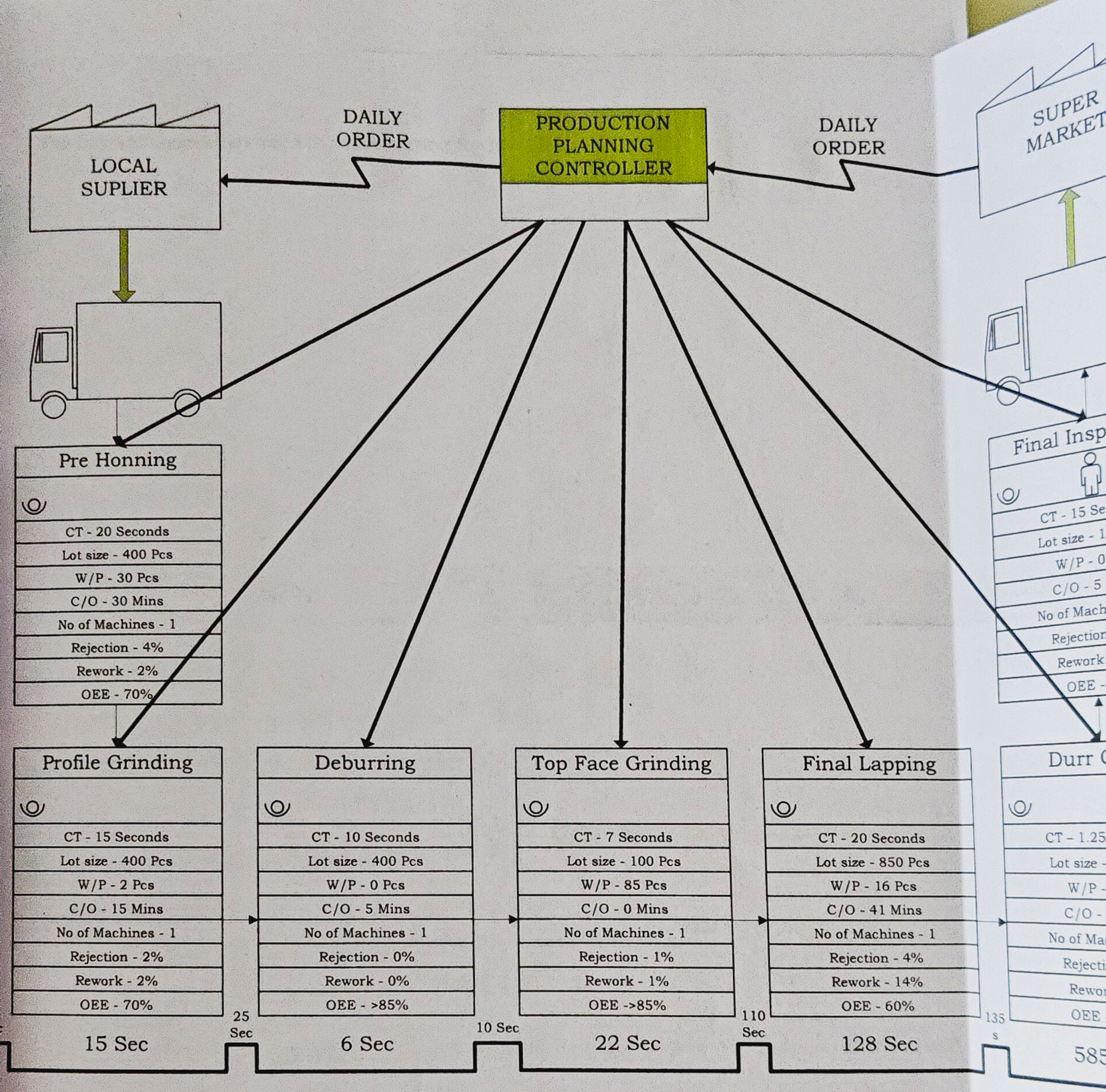


Space for Inserting the Print of the Chart (Leveraging Software Tool):



Key Learning and Inference:

* Swim lane charts clearly show who is responsible for each step.
* They assist in determining dependencies and handoffs the role of departments.
* When activities cross multiple lanes, process delays become apparent.
* The workflow redundant or superfluous steps are highlighted in the chart.
* It enhances comprehension of cross functional cooperation.
* Tracking task movement across lanes makes bottlenecks easier to identify.
* Swim lanes improve accountability and process transparency.
* They encourage improved process optimization and standardization.
* It is simple to identify communication gaps blue teams.
* Visual process mapping can increase overall workflow efficiency.



Parameters	Max. Marks	Marks Obtained	Signature of Staff in-charge
Conceptual Understanding	05		
Conduction of the experiment	06		
Analysis, Results and Inference	06		
Viva	03		
Total	20		

Inference:

* VSM helps to makes inefficiencies easily visible by visualizing the entire information and material flow.
* It draws attention to non-value added activities like needless movement, excess inventory and delays.
* By contrasting current state and future state process, VSM facilitates improved decision making.
* By cutting lead time and making the best use of available resource, the method increases productivity.
* In general VSM supports efforts for continous improvement and improves process transparency.

Parameters	Max. Marks	Marks Obtained	Signature of Staff in-charge
Conceptual Understanding	05		
Conduction of the experiment	06		
Analysis, Results and Inference	06		
Viva	03		
Total	20		

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OBSERVATION SHEET

Date: 19/11/2025 Name: Sudhanva Reg.No. RUCE25BIM405

Class: IEM WSO Expt No. WM01

Observation:-

Sl. No.	% Observed Rating (R_0)	Actual pace (x) in km/hr	Actual Rating	Name of the person walki
1	100%	4.8	100%	
2				
3	50%	6	125%	
4	20%	3.5	72.91%	
5	10%	1.5	31.25%	
6	10%	2.5	52.08%	
7	25%	4	83.33%	
8	30%	3	62.5%	
9				
10				

Signature of the Staff in-charge

Tabulation:

Sl.No.	% Observed rating R_0	Actual pace (x) Km/hr.	% Actual rating R_a	% Error	| % Error |
1.	100	4.8	100	0	0
2.	50	6	125	-150	150
3.	20	3.5	72.91	-264.55	264.55
4.	10	1.5	31.25	-212.5	212.5
5.	10	2.5	52.08	-420.8	420.8
6.	25	4	83.33	-233.32	233.32
7.	30	3	62.5	-108.33	108.33
8.					
9.					
10.					

Specimen Calculation:

3rd Reading.

① $R_a = \frac{x \times 100}{4.8} = \frac{3.5 \times 100}{4.8} = 72.91\%$

Where x = 2.75 km/hr

② $\%\ Error = \frac{R_0 - R_a}{R_0} \times 100$

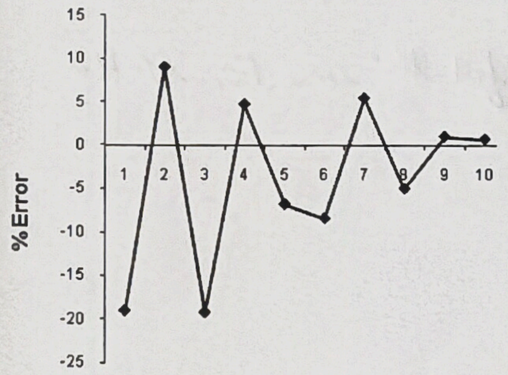
$= \frac{20 - 72.91}{20} \times 100$

$= -264.55$

$\simeq 264.55\%$

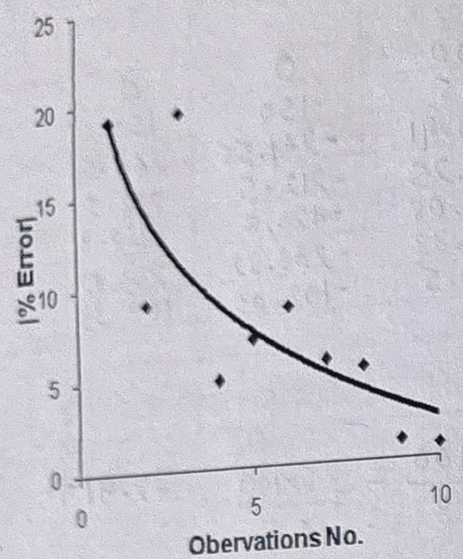
Graphs:

Learning Curve (Convergence Type)

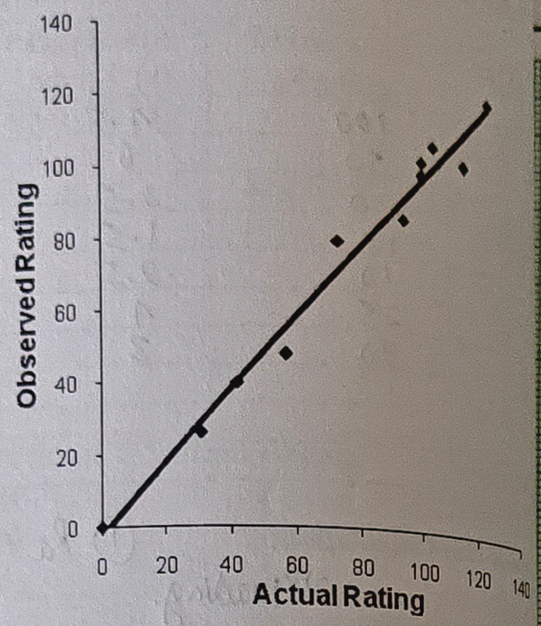


28.12

Learning Curve (Asymptotic Type)



Rating Calibration Curve



Inference:

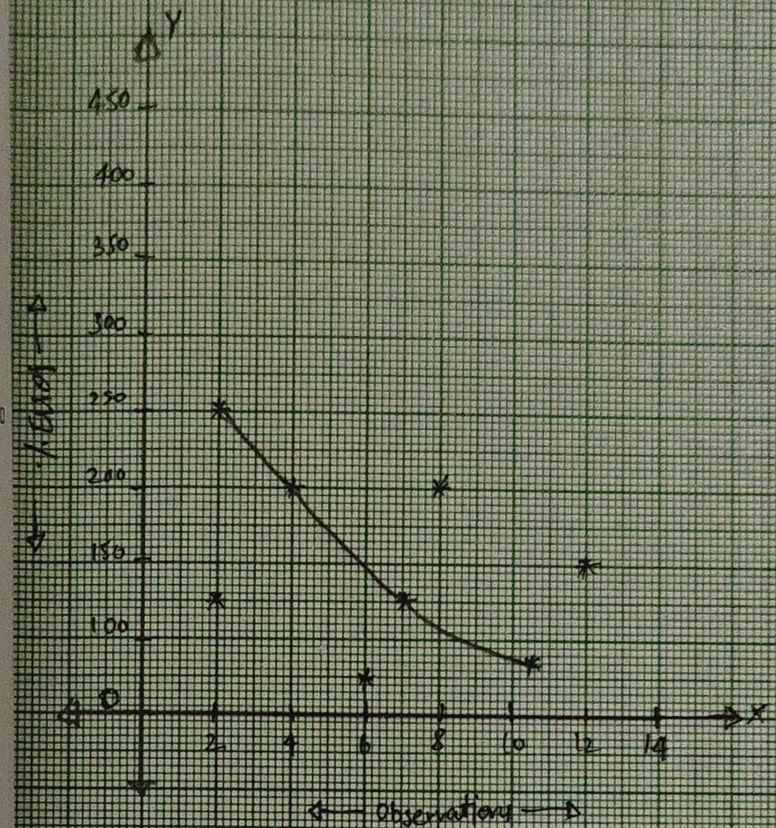
Here the Asymptotic curve depicts the ability of the observer to rate the activity of a person walking on the tread mill within the dlf of ±5% and the % of error lies almost blw -5 to 5.

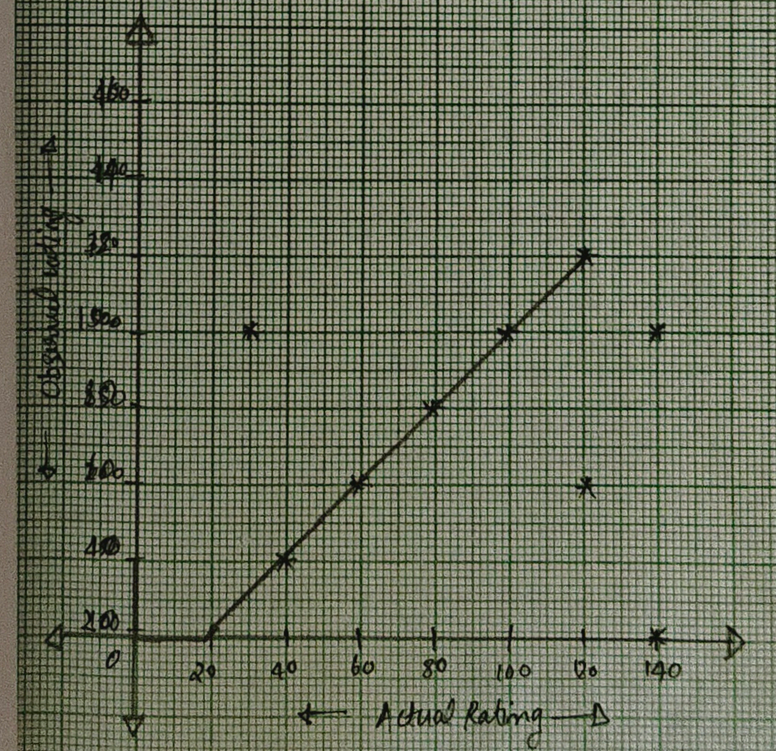
The calibration curve gives the output for the correction of the ratio of errors.

Date : 1 9 1 1 2 0 2 5

Sudhanva Jaisha
RVCE 25B1A0405

Scale:-
X-axis = 2 ⊡ observations
Y-axis = 1cm = 5%



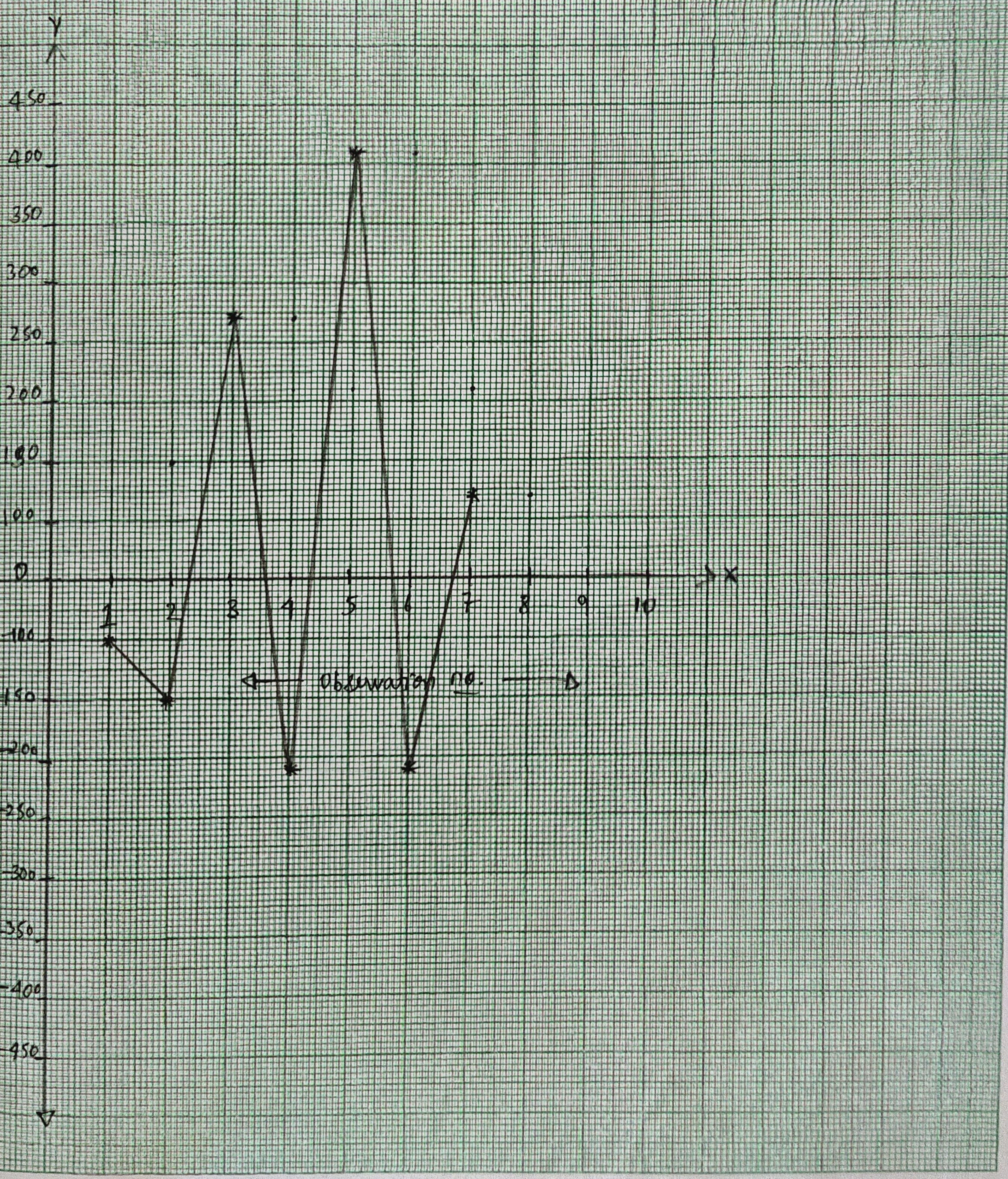


Date : 19/11/2025

Sumkarnika Vaishya
RVCE25BTI1424005

Learning Curve

Scale:-
X axis = 1cm = 1 slno
Y axis = 1cm = 5 Y-



Chandra's®

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Experiment: Advanced work measurement
Name: Sudhanva · Lalitha
Date: 15/10/25

Reg. No: RVCE25BIM405
Expt. NO. WM03

MTM ANALYSIS SHEET

Element	Time (in cms)	Code	Description
Reach	7.5	R18A	Reached from rest position to fixed position
Grasp	2	G1A	Grasped the marble from fixed position
Move	11.1	M18C	Move from fixed position by grasping marble to another fixed position.
Position	5.6	P1S	Position the marble on the fixed position.
Release	2.0	RL1	Release the marble on the fixed position.

Signature of Staff in-charge

Calculations:
Standard Time required for transferring 20 marbles from stand to marble collection b
simultaneously by both left and right hands.

$= x$ X 0.06 X 1.33 (Where x - total time of the activity)

$= 28.2 \times 0.06 \times 1.33$

$= 2.255036 \times 10$

$= 22.5036 \simeq 23$ seconds

Note:
1 TMU = 0.06 centiminutes
Indian standards scale is taken as 133

Inference:

* Reach -> grasp -> Move -> Position -> Release is the set order for this
  structured repetitive task.
* Each MTM elements time values are low, suggesting that the activity
  is easy, short cycle, and requires little physical effort.
* The method is effective for small scale manual handling.
* The marble stand and box are positioned to minimize reach
  distance, and the workplace layout is intended to minimize needless
  movement.

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NAME	Sudhanva. Seetha		
EXPERIMENT	FMBP	REG. NO	RVCE25BIM405
		DATE	10/12/2025

Observations:

- Pedal RPM: 60
- Rest BP: 120/84 mm Hg Pulse = 77
- Name of the subject: Shagas Chandra

Tabulation:

Sl. No	Time (min)	Load	Blood pressure in mm Hg	
			Systolic value	Diastolic value
1	2	0	133	73
2	2	20	128	~~77~~
3	2	40	130	79
4	2	60	136	78
5	2	80	145	76
6	2	100	137	73

Observations

- Pedal RPM: 60
- Rest BP: 109/55 mm Hg Pulse : 82
- Name of the subject: Tejas

Tabulation:

Sl. No	Time	Load	Blood pressure in mm Hg	
			Systolic value	Diastolic value
1	2	0	99	43
2	2	20	107	52
3	2	40	105	45
4	2	60	130	62
5	2	80	130	58
6	2	100	137	56

Observations

- Pedal RPM: 60
- Rest BP: 114/72 mm Hg Pulse = 92
- Name of the subject: Yashwanth.L

Tabulation:

Sl. No	Time	Load	Blood pressure in mm Hg	
			Systolic value	Diastolic value
1	2	0	117	79
2	2	20	118	78
3	2	40	129	80
4	2	60	129	76
5	2	80	124	74
6	2	100	132	78

Observations

- Pedal RPM: 60
- Rest BP: 120/80 mm Hg Pulse = 85
- Name of the subject: Sudhanva. Seitha

Tabulation:

Sl. No	Time	Load	Blood pressure in mm Hg	
			Systolic value	Diastolic value
1	2	0	120	80
2	2	20	126	82
3	2	40	132	84
4	2	60	138	86
5	2	80	144	88
6	2	100	150	90

Signature of the Staff in-charge

Inference:

* Progressive rise in systolic blood pressure with increasing load indicates increased cardiovascular effort; excessive rise or delayed recovery suggests onset of fatigue.
* Diastolic blood pressure shows minimal change or slight increase; an abnormal rise indicates peripheral resistance and muscular fatigue.
* Slow return of blood pressure to resting values after workload reflects reduced recovery capacity and confirms fatigue in the subject.

Parameters	Max. Marks	Marks Obtained	Signature of Staff in-charge
Conceptual Understanding	05		
Conduction of the experiment	06		
Analysis, Results and Inference	06		
Viva	03		
Total	20		

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NAME	Sudhanva. Smitha		
EXPERIMENT	SpO2 Fatigue measurement	REG. NO	1RV25IM405
		DATE	7/1/26

Observations

- Pedal RPM: 5
- Rest SpO_2: 98/68
- Name of the subject: Tejas

Tabulation:

Sl. No	Time in min	Load in Watts	Derived pulse rate value	SpO_2 value
1	2	1	122	98
2	2	2	128	94
3	2	3	138	99
4	2	4	142	93
5	2	5	144	95
6				

Observations

- Pedal RPM: 5
- Rest SpO_2: 97/145
- Name of the subject: Yashas

Tabulation:

Sl. No	Time in min	Load in Watts	Derived pulse rate value	SpO_2 value
1	2	1	142	99
2	2	2	152	99
3	2	3	145	98
4	2	4	157	98
5	2	5	159	98
6				

Observations

- Pedal RPM: 5
- Rest SpO_2:
- Name of the subject: Shreni

Tabulation:

Sl. No	Time in min	Load in Watts	Derived pulse rate value	SpO_2 value
1	2	1	164	99
2	2	2	156	99
3	2	3	122	100
4	2	4	138	99
5	2	5	140	99
6				

Observations

- Pedal RPM: 5
- Rest SpO_2: 98/82
- Name of the subject: Sneha

Tabulation:

Sl. No	Time in min	Load in Watts	Derived pulse rate value	SpO_2 value
1	2	1	103	98
2	2	2	128	98
3	2	3	136	99
4	2	4	124	99
5	2	5	141	99
6				

Signature of the Staff in-charge

Inference:

1) Normal in SpO_2 indicates fatigue when a person becomes fatigued the body oxygen demand increases. if SpO_2 level starts dropping from the range (95-100%) it reports reduced oxygen delivery to tissues, which is a sign of physical fatigue.

2) Prolonged low SpO_2 reflects sustained fatigue if SpO_2 remains lower than normal for a longer period it indicates that the body is not recovering efficiently..

3) SpO_2 recovery rate shows fatigue level. After activity if slow return of SpO_2 to normal level indicates higher fatigue further recovery of SpO_2 reports better physical endurance and lower fatigue.

Parameters	Max. Marks	Marks Obtained	Signature of Staff in-charge
Conceptual Understanding	05		
Conduction of the experiment	06		
Analysis, Results and Inference	06		
Viva	03		
Total	20		

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NAME	Sudhanva Sirsha	REG. NO	RUCE25BIM405
EXPERIMENT	APM - Standing	DATE	10/12/25
Observations	Subject :- Omkar . H . K		

Tabulation:

Sl. No.	Anthropometric Parameter	Dimensions in mm	Sl. No.	Anthropometric Parameter	Dimensions in mm
1.	Weight	56 Kg	17.	Arm reach from wall	860
2.	Stature	1660	18.	Elbow height	100
3.	Eye height	1470	19.	Wall to acromion distance	139
4.	Acromial height	1370	20.	Chest depth	200
5.	Metacarpal-III height	765	21.	Wall to lumbo sacral joint	60
6.	Knee height	420	22.	Abdominal extension to wall	230
7.	Biacromial breadth	460	23.	Bideltoid breath	520
8.	Chest breadth	350	24.	Inter seye breadth	420
9.	Hip breadth	420	25.	Waist breath	350
10.	Chest circumference	799	26.	Thumb tip reach	850
11.	Waist circumference	972	27.	Shoulder grip length	780
12.	Thigh circumference	460	28.	Span	1640
13.	Calf circumference	309	29.	Span akimbo	860
14.	Vertical reach	200	30.	Head breadth	210
15.	Vertical grip reach	203	31.	Head length	260
16.	Olecranon height	101	32.	Menton to top of the head	240

Signature of the Staff in-charge

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NAME	Sudhanva. Lolla		
EXPERIMENT	APM-Sitting	REG. NO	RVCE25BIM405
Observation: Subject :- Tejas NC		DATE	10/12/2025

Tabulation:

Sl. No.	Anthropometric Parameter	Dimensions in mm	Sl. No.	Anthropometric Parameter	Dimensions in mm
1.	Waist back length	420	22.	Hand length	183
2.	Sitting height	855	23.	Hand breadth at metacarpal –III	85
3.	Sitting eye height	740	24.	Hand breadth across thumb	102
4.	Popliteal height sitting	405	25.	Palm length	105
5.	Knee height sitting	555	26.	Hand thickness at metacarpal-III	27
6.	Thigh clearance height sitting	180	27.	First phalanx digit III length	66
7.	Elbow rest height	190	28.	Grip diameter (inside)	52
8.	Coroniod fossa to hand length	447	29.	Grip diameter (outside)	15
9.	Vertical grip reach height sitting	1260	30.	Middle finger palm grip diameter	47.5
10.	Sitting acromial height	580	31.	Grip spam	99
11.	Abdominal depth sitting	225	32.	Maximum grip length	246
12.	Buttock knee length	630	33.	Foot length	95
13.	Forearm hand length	440	34.	Foot breadth(ball of foot breadth)	90
14.	Buttock popliteal length	465	35.	Heel breadth	60
15.	Hip breadth sitting	305	36.	Instep length	181
16.	Elbow – elbow breadth	490	37.	Lateral malleolus height	95
17.	Knee-knee breadth	215	38.	Bimalleolar breadth	71
18.	Functional leg length	940	39.	Medial malleolus height	100
19.	Elbow grip length	380			

Signature of the Staff in-charge

Parameters	Max. Marks	Marks Obtained	Signature of Staff in-charge
Conceptual Understanding	05		
Conduction of the experiment	06		
Analysis, Results and Inference	06		
Viva	03		
Total	20		